PROJECT REPORT OF CO-OP Project at Industry (Module-2)
ON

TradeLearn

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

Submitted by:

Rajvardhan Tiwari (2210990710)

Supervised By:

Dr. Ajay Kumar

Associate professor

Chitkara University

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CHITKARA UNIVERSITY INSTITUTE OF ENGINEERING AND TECHNOLOGY
CHITKARA UNIVERSITY, PUNJAB, INDIA

CONTENTS

<u>DECLARATION</u>

I hereby certify that the work which is being presented in the project report entitled **“TradeLearn”** in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of **Ajay Katiyar**. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place: Rajpura **Rajvardhan(2210990710)**

Date: **31-04-2026**

This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

Dr. Ajay Kumar

Associate professor

Department of Computer Science and Engineering

Chitkara University Institute of Engineering and Technology,

Chitkara University, Punjab, India

ACKNOWLEDGEMENT

We express our sincere gratitude to our supervisor **Dr. Ajay Kumar**, Associate Professor, Department of Computer Science and Engineering, Chitkara University Institute of Engineering and Technology, for her continuous guidance, valuable suggestions, and constant encouragement throughout the development of this project.

We would also like to thank the faculty members of the Department of Computer Science and Engineering for providing us with the necessary resources and academic support required to complete this project successfully.

We are equally grateful to our friends and family members for their motivation and support during the course of this work.

Finally, we extend our appreciation to all those who directly or indirectly contributed to the successful completion of this project.

Rajvardhan

2210990710

ABSTRACT

The project titled **“TradeLearn: Learn Skills, Build Real Value”** aims to design and develop a comprehensive full-stack web application that enables users to exchange skills through a structured peer-to-peer learning system. The platform is designed to create an accessible and collaborative environment where individuals can teach skills they possess and learn new skills from others without relying on monetary transactions. The system introduces a credit-based learning model where knowledge itself acts as a valuable resource.

The primary objective of the project is to eliminate financial barriers associated with traditional learning platforms and provide users with a centralized platform for skill sharing, learning management, and collaborative education. The application allows users to create personalized profiles, add skills they can teach and skills they wish to learn, search for learning opportunities, and interact with other users through a structured exchange process.

The system incorporates secure authentication mechanisms using **JWT-based authentication**, protected routes, and session management to ensure user privacy, security, and reliable access control. Users can register accounts, manage profiles, send skill exchange requests, monitor activities, and maintain records of their learning interactions.

One of the major features of the platform is the **credit-based exchange mechanism**, where users earn credits by teaching skills to others and use those credits to learn new skills. This approach promotes active participation and creates a self-sustaining educational ecosystem that rewards knowledge sharing and contribution.

TradeLearn also implements a dashboard management system that enables users to track request history, available credits, ongoing learning activities, and exchange progress in real time. The application additionally provides search and browsing functionality to help users identify suitable learning partners based on offered skills and interests.

The project follows a full-stack architecture using **HTML, CSS, EJS, Node.js, Express.js, MongoDB, and JWT authentication**, ensuring scalability, maintainability, and efficient client-server interaction.

1. Introduction

Overview of the Project

TradeLearn – Learn Skills, Build Real Value is a web-based peer-to-peer skill exchange platform designed to simplify and modernize the process of learning and knowledge sharing. In traditional learning environments, individuals often depend on paid educational platforms or one-way learning systems, which can create financial barriers and limit opportunities for collaboration. Many people possess valuable skills and expertise but lack a structured system through which they can exchange their knowledge with others.

The proposed system provides a comprehensive digital platform that integrates skill management, profile creation, request handling, credit tracking, and collaborative learning into a single unified application. By leveraging modern full-stack web technologies, the platform allows users to teach skills they possess and learn new skills from other users through a structured credit-based mechanism rather than monetary transactions.

The system centralizes key functionalities such as user registration, profile management, skill listing, skill discovery, exchange request handling, and dashboard tracking. Users can create profiles, specify the skills they can offer and the skills they want to learn, search for suitable learning partners, and participate in structured skill exchange activities. This centralized approach improves accessibility, encourages knowledge sharing, and creates a more organized learning experience.

Additionally, the application is designed with a responsive and user-friendly interface to ensure smooth interaction and ease of use for users across different devices and varying technical skill levels. Security mechanisms such as JWT-based authentication and protected access controls ensure reliable and secure platform functionality.

Overall, TradeLearn contributes to the digital transformation of collaborative learning by creating an accessible, community-driven ecosystem where knowledge acts as a form of currency. The platform promotes active participation, reduces financial dependency, and enhances learning opportunities through structured peer-to-peer interaction and knowledge exchange.

Problem Statement

Despite the availability of numerous online learning platforms and educational resources, many individuals still face difficulties in accessing quality learning opportunities due to financial limitations and lack of collaborative systems. Most existing platforms follow a traditional one-way learning approach where users only consume content through paid courses and subscriptions. These systems often limit participation and fail to create an environment where users can actively contribute their knowledge.

Another major challenge is the absence of a structured platform that allows individuals to exchange skills directly with one another. Many users possess valuable expertise in areas such as programming, communication, design, academics, and technical skills, but there are limited opportunities to utilize this knowledge for mutual benefit. Existing systems generally do not provide mechanisms for recognizing teaching contributions unless monetary transactions are involved.

Additionally, users often face difficulties in discovering suitable learning partners, managing learning requests, tracking exchange activities, and maintaining records of their progress. The lack of a centralized and organized system reduces engagement and limits the effectiveness of peer-to-peer learning experiences.

Therefore, there is a strong need for a secure, scalable, and community-driven skill exchange platform that enables users to both teach and learn through a structured credit-based mechanism. The system should support profile management, skill discovery, request handling, activity tracking, and secure authentication while creating an interactive ecosystem where knowledge itself acts as a form of currency. TradeLearn aims to address these challenges by promoting collaborative learning and making skill development more accessible without relying on monetary transactions.

Objectives of the Project

The primary objective of **TradeLearn** is to develop a reliable and collaborative skill exchange platform that simplifies and promotes peer-to-peer learning through a structured credit-based system. The system is designed with the following specific objectives:

● To create a centralized platform for managing user profiles, skills, learning requests, credit transactions, and activity tracking efficiently

● To eliminate financial barriers associated with traditional learning platforms by introducing a knowledge-sharing model based on credits instead of monetary transactions

● To implement secure authentication mechanisms using modern technologies such as JWT authentication and protected routes to ensure user security and data privacy

● To enable users to add and manage skills they can teach and skills they want to learn in an organized manner

● To provide a structured skill exchange mechanism where users can send, accept, reject, and manage learning requests efficiently

● To implement a credit-based system where users earn credits by teaching others and use credits to learn new skills

● To provide dashboard functionality for tracking credits, request history, ongoing learning activities, and exchange progress in real time

● To design a user-friendly and responsive interface that improves usability and accessibility across different devices

● To encourage collaborative and community-driven learning where users actively participate as both contributors and learners

● To ensure scalability of the system so that additional features such as real-time communication, AI-based recommendations, certifications, and mobile applications can be integrated in the future

These objectives collectively aim to create an accessible, organized, and interactive learning ecosystem that promotes knowledge sharing, encourages community participation, and enhances skill development through collaborative education.

Scope of the Project

The scope of **TradeLearn** includes the design and development of a fully functional web-based peer-to-peer skill exchange platform that addresses the core learning and knowledge-sharing needs of users. The system primarily focuses on enabling users to teach and learn skills through a structured credit-based mechanism while promoting collaboration and community-driven learning.

The application allows users to register, log in, manage their profiles, and specify skills they can offer as well as skills they wish to learn. Users can browse available skills, search for suitable learning partners, and send learning requests to other users. The platform also provides functionality for managing requests, tracking exchange activities, and maintaining learning records efficiently.

The system incorporates a credit-based exchange mechanism where users earn credits by teaching others and use earned credits to learn new skills. Additional features such as dashboard management, request tracking, learning history, and activity monitoring help users manage their interactions and progress effectively.

TradeLearn also provides secure authentication mechanisms and protected user access to ensure privacy and reliable system operation. The platform supports profile management, skill discovery, and organized exchange workflows to create a seamless learning experience.

While the current system includes all essential functionalities required for skill exchange and collaborative learning, it has been designed with scalability in mind. Future enhancements may include AI-based skill recommendations, real-time chat and video communication, mobile application support, group learning communities, certification systems, and advanced analytics dashboards.

The project is currently implemented as a web-based application; however, it can be extended to mobile platforms to improve accessibility and user engagement. Additional integrations with external learning resources and communication services can further enhance platform capabilities.

In summary, the scope of this project is to establish a strong foundation for a modern and scalable collaborative learning ecosystem that supports continuous growth, promotes knowledge sharing, and adapts to future technological advancements and user requirements.

2. Methodology

2.1 Development Approach

The development of TradeLearn follows the Agile software development methodology, which focuses on iterative development, flexibility, and continuous improvement. Unlike traditional development approaches, Agile allows features to be developed and improved through multiple development cycles based on testing and feedback.

The project was divided into smaller modules, and each module was designed, implemented, tested, and refined independently before integration. Regular development and testing helped identify issues early, improve platform functionality, optimize workflows, and enhance overall system performance.

Agile methodology also supported effective task distribution and parallel development of major modules such as authentication, profile management, skill exchange systems, dashboard implementation, database integration, and frontend interface development.

2.2 Requirement Analysis

Requirement analysis is an important phase in system development where user needs and expectations are identified and documented. In this project, requirements were determined by analyzing common limitations in existing learning systems, such as financial barriers, lack of collaboration, and absence of structured peer-to-peer learning.

The requirements were categorized into two main types:

2.2.1 Functional Requirements

These define the core functionalities of the system, including:

- User registration and secure authentication
- Profile management and skill listing
- Skill exchange request handling
- Credit-based learning system
- Search and browse functionality
- Dashboard and activity monitoring
- Request tracking and learning history management

2.2.2 Non-Functional Requirements

These focus on performance and system quality:

- Secure authentication and user data protection using JWT
- Scalability for increasing users and future feature expansion
- User-friendly and responsive interface across devices
- Reliability and availability of platform services
- Efficient database handling and optimized performance

2.3 System Design

The system design phase involved planning the architecture and structure of the application to ensure scalability, maintainability, security, and smooth integration between system modules.

2.3.1 Architecture Design

TradeLearn follows a **client-server architecture**:

- The frontend handles user interaction, dashboards, and profile management
- The backend processes authentication, business logic, credit calculations, request handling, and API processing
- MongoDB stores user information, skills, requests, and credit data

2.3.2 Database Design

A structured database schema was designed to store and manage data related to users, profiles, skills, exchange requests, credits, and activity records. Proper relationships and data structures were established to maintain consistency and avoid redundancy.

2.3.3 User Interface Design

The user interface was designed to be responsive, intuitive, and easy to use. The application includes profile pages, dashboards, request management interfaces, search functionality, and activity tracking sections.

Wireframes and layouts were planned before implementation to maintain consistency and improve user experience.

2.4 Development Process

The development phase involved implementing all designed modules using full-stack web technologies.

2.4.1 Frontend Development

The frontend was developed using HTML, CSS, and EJS, enabling dynamic rendering and interactive user interfaces. The design focuses on simplicity, responsiveness, and ease of navigation.

2.4.2 Backend Development

The backend was developed using Node.js and Express.js, which manage server-side logic, route handling, authentication, request processing, and communication with the database.

RESTful APIs were implemented for efficient interaction between frontend and backend systems.

2.4.3 Database Integration

MongoDB was used as the primary database to store application data efficiently. MongoDB enables flexible schema design and seamless integration with JavaScript technologies.

2.4.4 Authentication and Authorization

JWT (JSON Web Tokens) was implemented for authentication and session management. Protected routes and middleware were used to ensure secure user access and data protection.

2.5 Testing and Deployment

Testing ensures application reliability, performance, and usability.

2.5.1 Testing

• **Unit Testing:** Individual modules such as authentication, profile management, request handling, and APIs were tested independently.

• **Integration Testing:** Communication between frontend, backend, and database systems was verified.

• **User Testing:** The application was tested from the user's perspective to improve usability, responsiveness, and overall experience.

2.5.2 Deployment

After successful testing, the application can be deployed using cloud platforms:

• Frontend and backend deployment through cloud hosting platforms such as Vercel or Render

• MongoDB Atlas used for cloud database management

• Proper environment configuration and security setup performed before deployment

Deployment ensures online accessibility, scalability, and real-world usability of the platform.

3 Tools and Technologies

3.1 Frontend

The frontend of **TradeLearn** is developed using **HTML, CSS, and EJS (Embedded JavaScript Templates)**. HTML is used to create the structural layout of the application, while CSS is used for styling and creating responsive user interfaces. EJS enables dynamic server-side rendering and allows application data to be displayed efficiently on web pages.

The frontend is designed to provide a clean, interactive, and user-friendly experience for users. Features such as user registration, profile management, dashboard viewing, skill browsing, and request handling are integrated into the interface. The responsive design ensures compatibility across different devices and screen sizes, improving accessibility and user experience.

3.2 Backend

The backend of TradeLearn is implemented using **Node.js and Express.js**, which provide a scalable and efficient environment for handling server-side operations, authentication, request processing, business logic, and RESTful API development. Express.js simplifies route handling and communication between frontend and backend components.

The backend manages core functionalities such as user authentication, profile management, skill exchange requests, dashboard operations, and credit calculations. It processes user interactions efficiently and ensures smooth communication between the application layers.

Additionally, **JWT (JSON Web Token)** authentication is integrated to provide secure login functionality, session management, and protected route access. Middleware functions are implemented to improve security and ensure controlled access to application resources.

3.3 Database

MongoDB is used as the primary database for TradeLearn. MongoDB is a document-oriented NoSQL database that provides flexibility, scalability, and efficient data storage for modern web applications.

The database stores important information such as user profiles, offered skills, learning requests, credit transactions, authentication details, and activity records. MongoDB ensures reliable data handling and seamless integration with the JavaScript-based technology stack used in the application. It also supports future scalability requirements and efficient data retrieval operations.

4 Implementation

4.1 Architecture

The Smart Study Planner system is designed using a client-server architecture, which ensures The TradeLearn system is designed using a **client-server architecture**, which ensures clear separation between the user interface and backend processing logic. This architecture improves scalability, maintainability, performance, and efficient handling of skill exchange operations.

In this model, the frontend (client-side), developed using **HTML, CSS, and EJS**, manages user interaction, profile interfaces, dashboard rendering, skill browsing, and responsive user components. The frontend communicates with the backend through secure **RESTful APIs** for seamless data exchange and smooth user interaction.

The backend, developed using **Node.js and Express.js**, processes API requests, authentication, business logic, request handling, credit calculations, and skill exchange workflows. It also manages user activities, request tracking, profile management, and dashboard functionalities.

API communication plays a major role in the system by enabling efficient interaction between frontend and backend modules. Operations such as user registration, authentication, profile updates, skill management, request handling, and dashboard updates are processed through secure API endpoints.

MongoDB is used as the primary database for storing user profiles, skills, request information, credit records, and activity history. **JWT authentication** is integrated to provide secure login sessions and protected access to application resources.

The overall architecture ensures that the system remains modular, secure, scalable, and flexible enough for future enhancements such as mobile applications, AI-based recommendations, real-time communication, and additional learning features.

4.2 Features

The TradeLearn system incorporates a comprehensive set of features designed to simplify skill exchange, promote collaborative learning, and provide a secure and engaging user experience.

● Secure **JWT-based authentication system** with login session management and protected routes to ensure secure user access and data privacy.

● User registration and profile management functionality that allows users to create accounts and maintain personalized profiles.

● Skill management system that enables users to add skills they can teach and skills they wish to learn.

● Skill exchange request system that allows users to send, accept, reject, and manage learning requests.

● Credit-based learning mechanism where users earn credits by teaching others and spend credits while learning new skills.

● Search and browse functionality that helps users discover available skills and suitable learning partners.

● Dashboard management system that displays request history, available credits, learning activities, and ongoing exchange progress.

● Request tracking system that monitors request states such as pending, accepted, rejected, and completed.

● Secure API communication between frontend and backend for efficient data processing and interaction.

● Responsive and user-friendly interface designed for accessibility across different devices and screen sizes.

● Scalable architecture that supports future enhancements such as real-time chat, AI recommendations, mobile support, and community learning features.

5 Results

5.1 Efficiency

The implementation of **TradeLearn** has significantly improved the efficiency of skill sharing and collaborative learning. By automating processes such as skill discovery, request handling, credit management, and activity tracking, the system reduces dependency on traditional and unorganized learning methods. This enables users to focus more on learning and teaching rather than manually managing interactions and communication.

The structured credit-based learning mechanism encourages active participation and creates an organized exchange environment. Features such as dashboards, request tracking, and skill search functionality further improve user experience and simplify the overall learning process. The platform also minimizes the barriers associated with traditional educational systems and promotes effective peer-to-peer learning.

5.2 Data Management

The system provides a centralized database for storing and managing all platform-related information, including user profiles, skills, exchange requests, credit transactions, authentication details, and activity records. This centralized approach reduces redundancy and ensures consistency across the application.

Users can easily access and update their information while tracking learning activities and exchange progress in real time. **MongoDB** provides flexible and efficient data storage, ensuring scalability and reliable information retrieval. Proper data organization and structured storage also contribute to better system performance and maintainability.

5.3 Security

Security is a critical component of TradeLearn, and the system ensures strong protection of user information through secure authentication and authorization mechanisms. **JWT-based authentication**, protected routes, and secure session management ensure that only authorized users can access platform features and personal information.

Additional security measures such as password protection, middleware validation, and controlled API access help maintain user privacy and preserve data integrity. Overall, the system provides a secure and reliable environment for users to participate in learning activities and exchange knowledge safely.

6 Future Scope

The **TradeLearn** system provides a strong foundation for collaborative skill exchange and peer-to-peer learning; however, several opportunities exist for further enhancement and expansion to make the platform more intelligent, interactive, and scalable.

● **AI-based skill recommendation system** to provide personalized suggestions for skills, learning partners, and learning paths based on user interests and activity patterns.

● **Mobile application development** with Android and iOS support for improved accessibility and learning on the go.

● **Real-time chat and video call integration** to enable direct communication and live learning sessions between users.

● **Notification and reminder system** for exchange requests, learning sessions, credit updates, and platform activities.

● **Collaborative community features** such as discussion forums, learning groups, and group skill-sharing sessions.

● **Advanced analytics dashboard** to provide insights regarding learning progress, credits earned, skill growth, and user activity trends.

● **Skill verification and certification mechanisms** for validating completed learning activities and increasing user credibility.

● **Gamification features** such as badges, achievement levels, rewards, and learning streaks to increase engagement and motivation.

● **Integration with external learning platforms** and educational resources to provide additional learning materials.

● **AI-powered learning assistance** capable of suggesting personalized learning strategies and optimized skill development paths.

These future enhancements can transform TradeLearn into a more intelligent, scalable, and comprehensive collaborative learning ecosystem capable of supporting broader educational and technological requirements.

7 Conclusion

TradeLearn – Learn Skills, Build Real Value successfully demonstrates how modern web technologies and collaborative learning concepts can transform traditional skill development methods into an interactive, accessible, and community-driven learning experience. The system effectively addresses the limitations of conventional educational platforms by providing a centralized, secure, and structured platform for skill exchange, request management, activity tracking, and peer-to-peer learning.

Through the implementation of modern full-stack technologies such as **HTML, CSS, EJS, Node.js, Express.js, MongoDB, and JWT authentication**, the project achieves scalability, reliability, secure user interaction, and efficient data management. Core functionalities such as profile management, skill discovery, exchange request handling, dashboard tracking, and credit-based learning significantly improve accessibility and encourage active participation among users.

The system eliminates financial barriers associated with traditional learning platforms by introducing a unique **credit-based knowledge exchange mechanism**, where users earn credits through teaching and utilize them for learning new skills. This approach promotes knowledge sharing, community participation, and continuous learning while creating a self-sustaining educational ecosystem.

TradeLearn reduces dependency on one-way content consumption and encourages users to become contributors as well as learners. The platform simplifies the process of discovering learning opportunities, managing interactions, and maintaining structured learning records through an organized workflow and intuitive user interface.

This project also demonstrates the successful integration of frontend development, backend services, database management, authentication mechanisms, and API communication into a complete full-stack application. The modular architecture and structured development approach contributed significantly to building a scalable and maintainable system.

In conclusion, TradeLearn provides a practical, scalable, and innovative solution for collaborative learning and skill development. It establishes a strong foundation for future enhancements and represents an important step toward creating a knowledge-sharing ecosystem where learning becomes more accessible, interactive, and community-driven.

8 References

8.1 Node.js Documentation. Available at: Node.js Official Documentation

8.2 Express.js Documentation. Available at: Express.js Documentation

8.3 MongoDB Documentation. Available at: MongoDB Documentation

8.4 JSON Web Tokens (JWT). Introduction and Specifications. Available at: JWT Documentation

8.5 EJS Documentation. Available at: EJS Official Documentation

8.6 GitHub Documentation. Available at: GitHub Documentation

8.7 Postman Documentation. Available at: Postman Learning Center

8.8 MongoDB Atlas Documentation. Available at: MongoDB Atlas Documentation

9 Appendices

Appendix A: System Screenshots

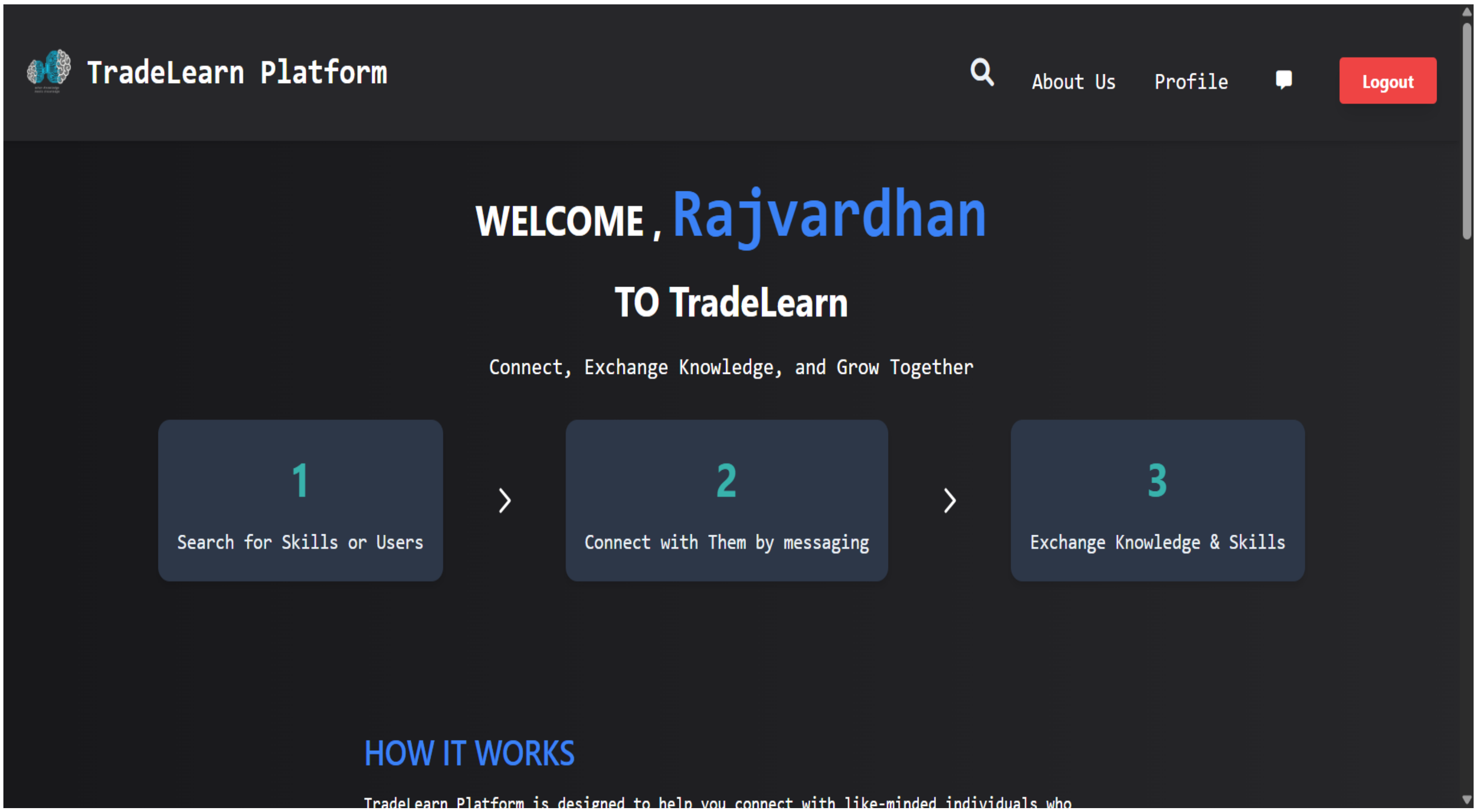
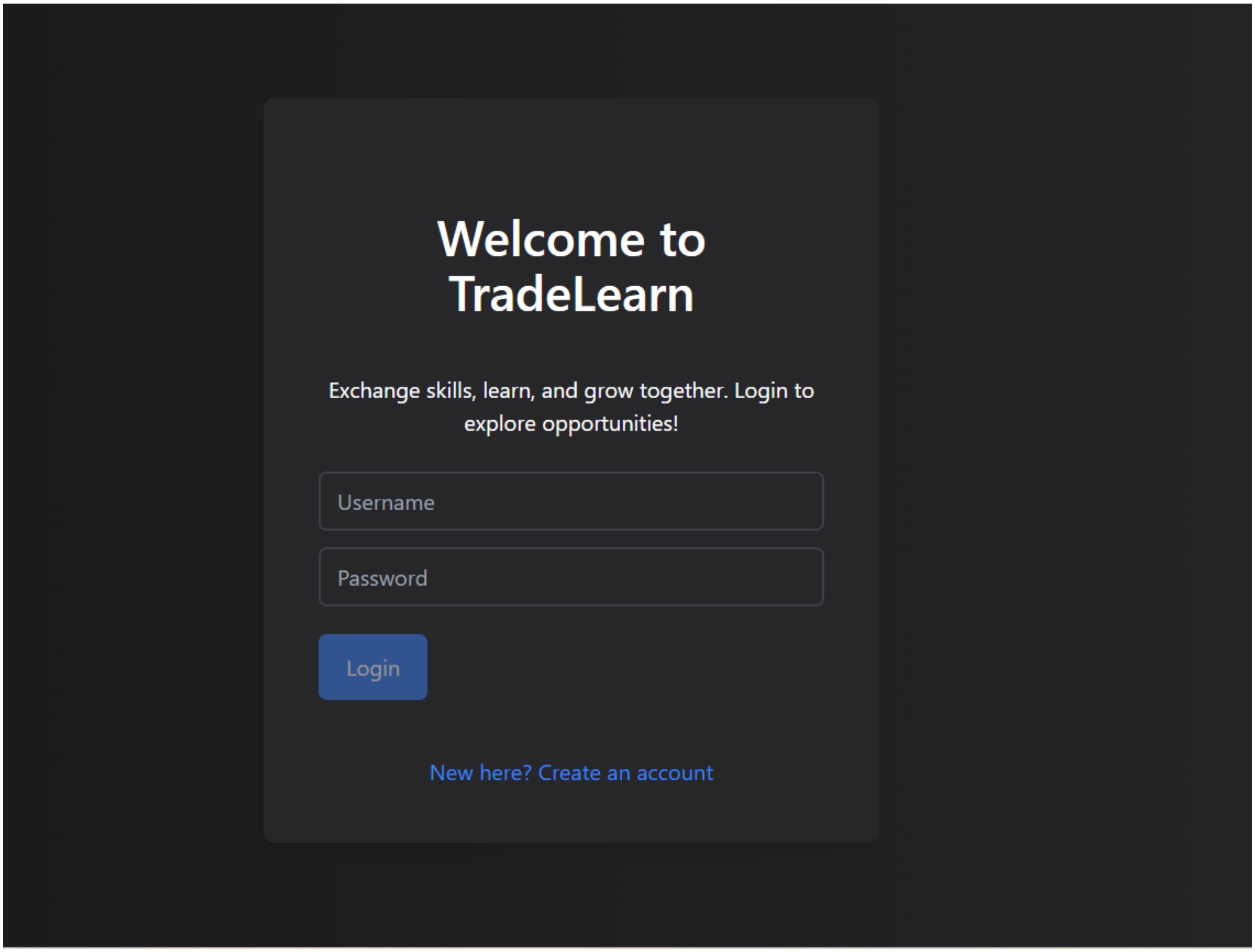
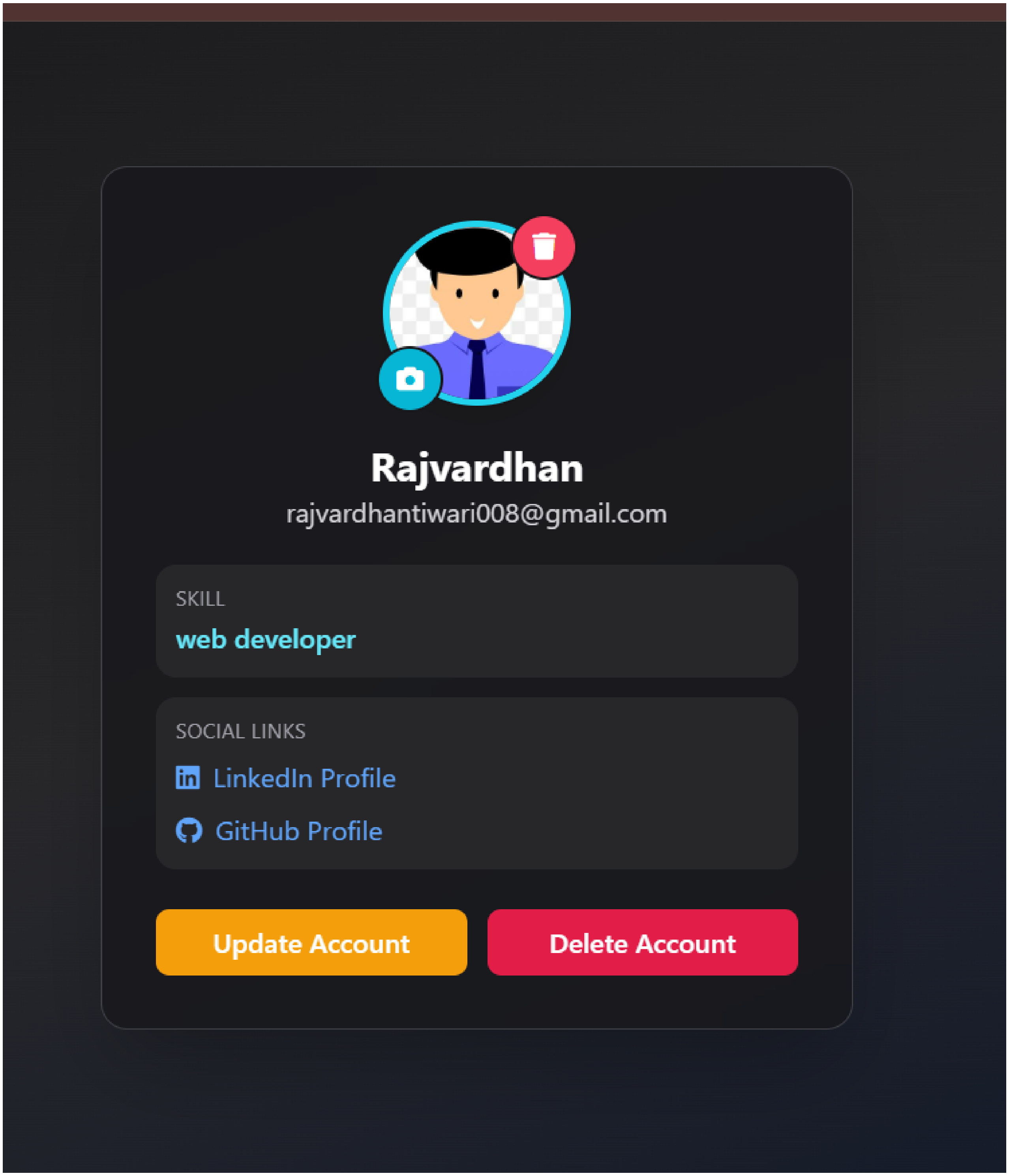


Figure 1- Dashboard

Figure 2 – Login Page





.

Figure 3- User Profile

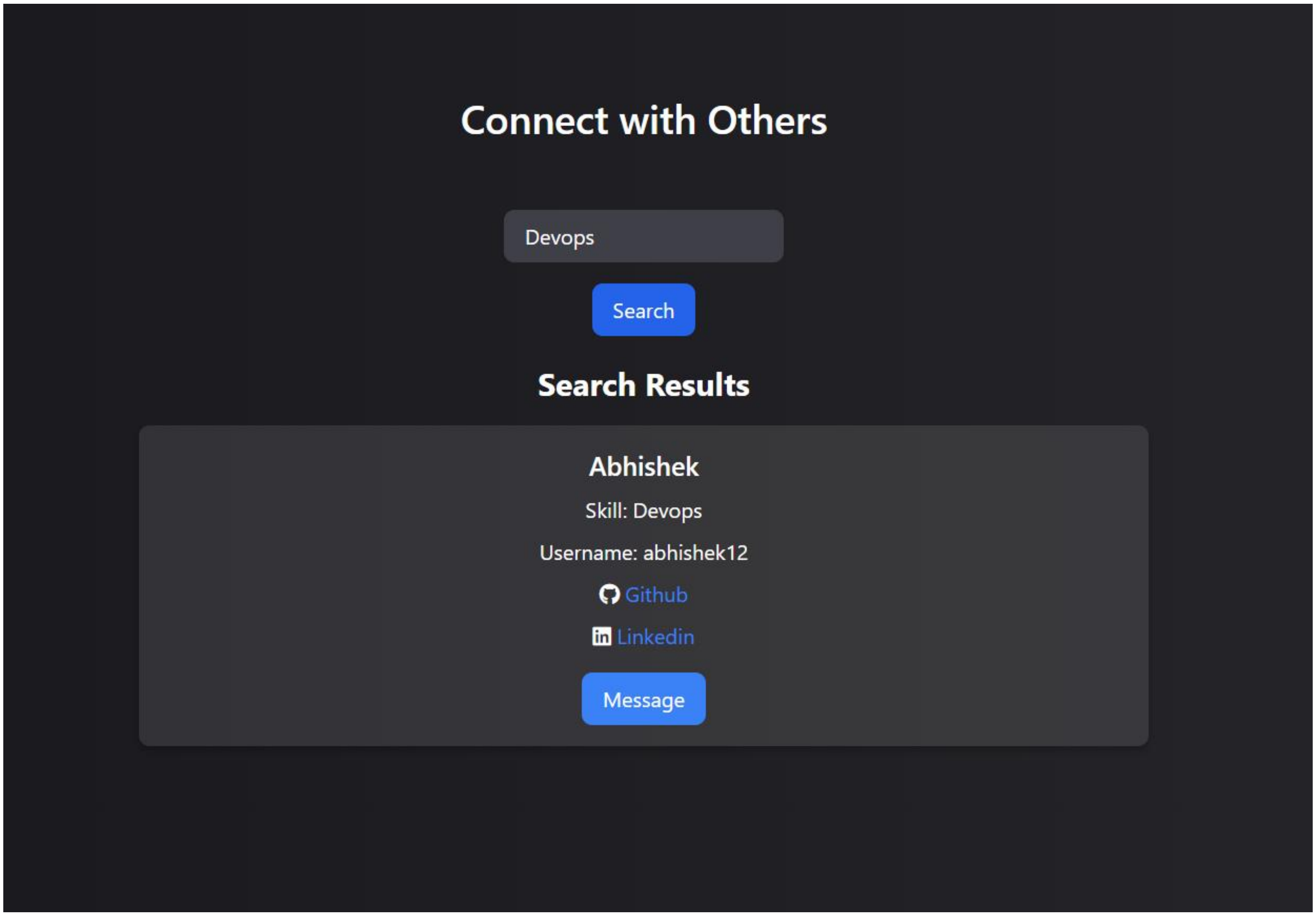


Figure 4- Mutlple user with there skills

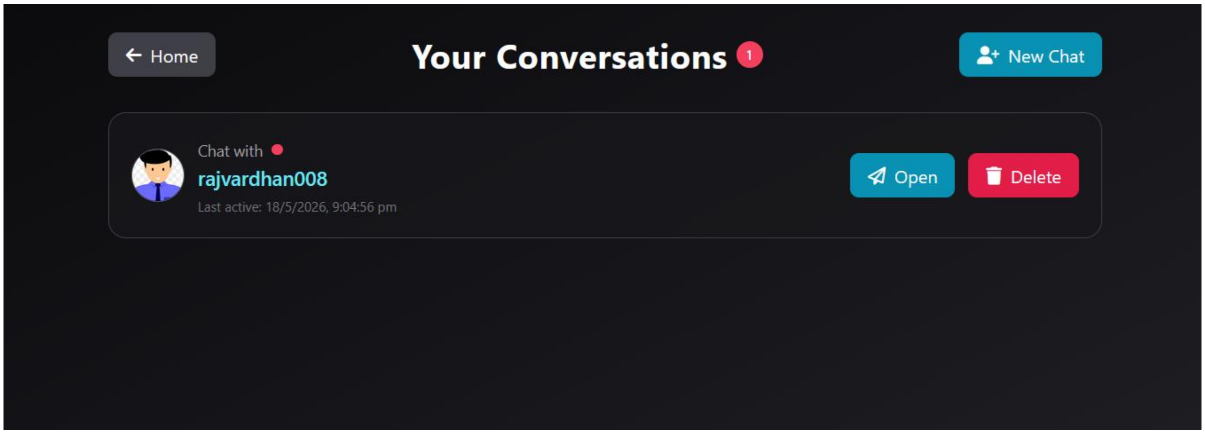


Figure 5.1: Chat Section

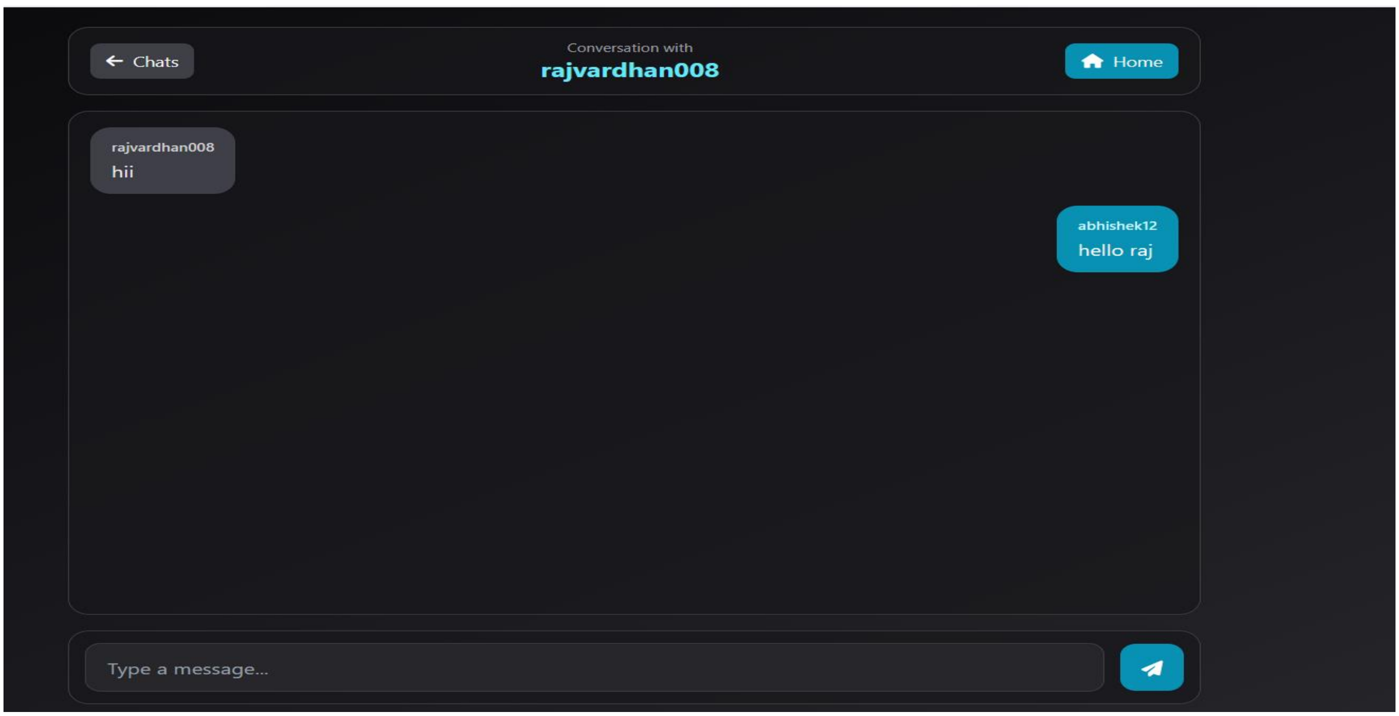


Figure 5.2 – chat with another user